1) Define ohm's law & write it's limitation

A) ohm's law :- ohm's law states that the current passing through a conductor is directly proportional to the voltage (V) & inversely proportional to the resistance at constant temperature.

$I \propto V$; $I = \frac{V}{R}$ (or) $V = IR$

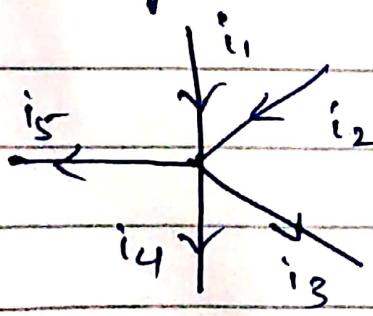
limitations :-

* It doesnot apply to semiconductors.
* The relations b/w voltage & current depends on sign of voltage
* It is NOT applicable to non-linear elements.

2) Explain kirchoff's laws with examples

A) KCL :- The sum of currents entering into a junction equals to the sum of currents flowing out of the junction.

Eg :-



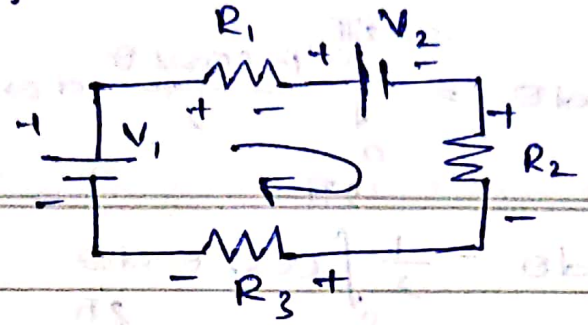
$\rightarrow i_1 + i_2 = i_3 + i_4 + i_5$

$i_1 + i_2 - i_3 - i_4 - i_5 = 0$

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KVL:- The algebraic sum of voltage drop in any closed circuit is zero.

Eg:-



$$-V_1 + iR_1 + V_2 + iR_2 + iR_3 = 0$$

$$V_1 = i(R_1 + R_2 + R_3) + V_2$$

5) Derive equations for RMS Value, Average value, Peak value, Peak factor, form factor for sinusoidal waveform.

A) RMS:- ~~The RMS value of an alternating current is defined as the square root of the mean of the squares~~

The RMS value is an measure of the effective value of an alternating quantity. It represents DC equivalent that would produce the same heating effect in a resistor as the AC does over a complete cycle.

$$I_{rms} = \sqrt{\frac{1}{T}\int_0^T i^2 d\theta}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi} I_m^2 \sin^2\theta \, d\theta}$$

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$$= \sqrt{\frac{I^2 m}{2\pi} \int_0^{2\pi} \sin^2\theta \, d\theta}$$

$$\left[\int_0^{2\pi} \sin^2\theta \, d\theta = \int_0^{2\pi} \frac{1-\cos 2\theta}{2} \, d\theta \right.$$

$$= \frac{1}{2}\int_0^{2\pi} 1 \, d\theta - \frac{1}{2}\int_0^{2\pi} \cos 2\theta \, d\theta$$

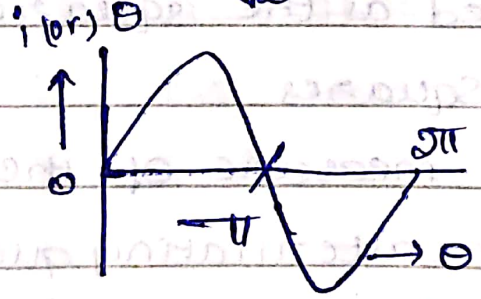
$$= \frac{1}{2} \cdot 2\pi - \frac{1}{2}\left[\frac{\sin 2\theta}{2}\right]_0^{2\pi}$$

$$= \pi - 0$$

$$= \pi \Big]$$

$$= \sqrt{\frac{I^2 m}{2\pi} \cdot \pi}$$

$$I_{rms} = \frac{I_m^2}{\sqrt{2}} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$



$$i = I_m \sin\theta$$

Average value :- The avg value of A.C is the arithmetic sum of all instantaneous values taken over a 1/2 cycle. Over a full cycle the value is '0'.

$$I_{avg} = \frac{1}{T}\int_0^T i \, d\theta$$

$$= \frac{1}{\pi}\int_0^{\pi} I_m \sin\theta \, d\theta$$

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$$= \frac{I_m}{\pi} \int_0^{\pi} \sin\theta \, d\theta$$

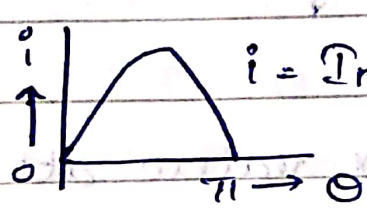
$$= \frac{I_m}{\pi} [-\cos\theta]_0^{\pi}$$

$$= \frac{I_m}{\pi} [-\cos\pi + \cos 0]$$

$$= \frac{I_m}{\pi} \times 2$$

$$I_{avg} = \frac{2I_m}{\pi} = 0.637 I_m$$

$$V_{avg} = 0.637 V_m$$



Peak factor:- It is defined as the ratio of peak value to the RMS value of wave

$$\text{Peak factor} = \frac{\text{Peak value}}{\text{RMS value}}$$

$$= \frac{I_m}{I_{rms}} = \frac{I_m}{0.707 I_m} = \sqrt{2} \text{ (or) } 1.414$$

Form factor:- It is defined as the ratio of RMS value to the avg value of the wave

$$\text{Form factor} = \frac{\text{RMS value}}{\text{Avg value}}$$

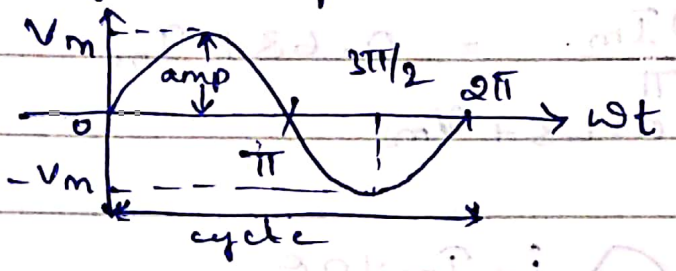
$$= \frac{0.707 I_m}{0.637 I_m} = 1.11$$

6) Explain the following terms

a) cycle b) Amplitude c) Time d) Frequency Period

e) Instantaneous value f) Peak to peak value

a) **Cycle:-** One complete set of positive & negative values of an alternating quantity is known as a cycle



Amplitude:- The max. value attained by an alternating quantity is called as amplitude.

Timeperiod:- The time taken in seconds to complete one cycle of an alternating quantity is called its timeperiod.

$$T = 1/f$$

Frequency:- The no. of cycles that occur in one second is called the frequency.

Instantaneous value:- The value of an alternating quantity at any instant is called instantaneous value.

Peak to Peak value:- The peak to peak value of sine-wave is the value from the positive to the negative peak.

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7) Explain the power triangle & importance of power factor.

A) Power triangle:-

This is a right angled triangle & indicates the relation among apparent power, true power & reactive power. It reveals the following facts about circuit:

* Power factor, $\cos\phi = \frac{\text{True power}}{\text{apparent power}} = \frac{VI\cos\phi}{VI}$

* $(\text{Apparent power})^2 = (\text{True power})^2 + (\text{reactive power})^2$

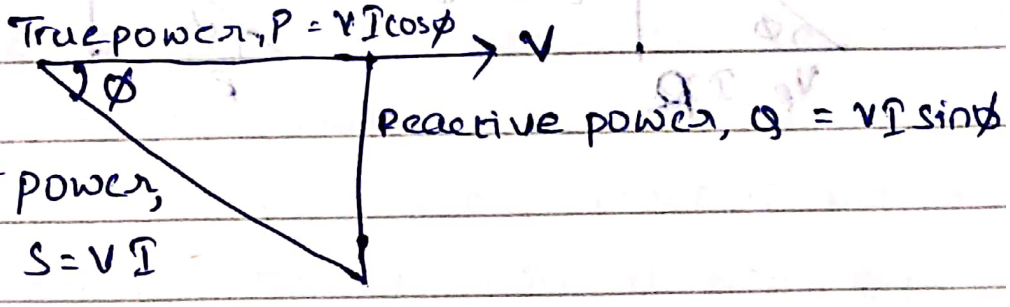
$$S^2 = P^2 + Q^2$$

* True power, P = Apparent power $\times \cos\phi$

$= VI\cos\phi$

Reactive power, Q = Apparent power $\times \sin\phi$

$= VI\sin\phi$



Power factor:- The power factor of a circuit can be defined as $\cos\phi = \frac{\text{Reactive. P}}{\text{apparent. P}}$

Importance:-

* It reduces power losses in the system
* It maintains voltage levels at load end, which leads to better equipment performance
* It improves efficiency.

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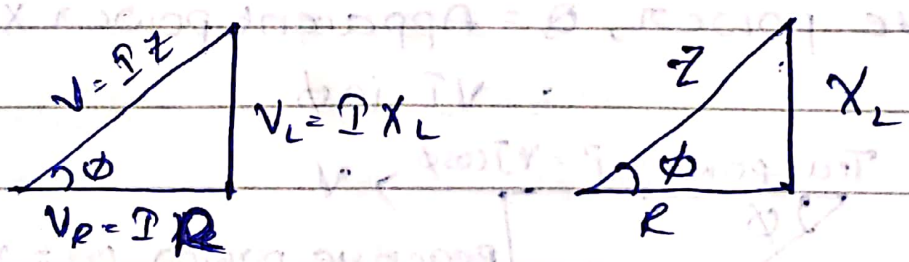
8) What is impedance & explain the impedance triangle.

A) Impedance :- It is the opposition to the current flow in A.C. It combines resistance & reactance.

$$\boxed{Z = R + jX}$$

→ Impedance triangle enables us to calculate

* impedance of circuit, $Z = \sqrt{R^2 + X_L^2}$
* power factor of circuit, $\cos\phi = R/Z$
* phase angle, $\tan\phi = X_L/R$
* Whether currents leads or lags voltage



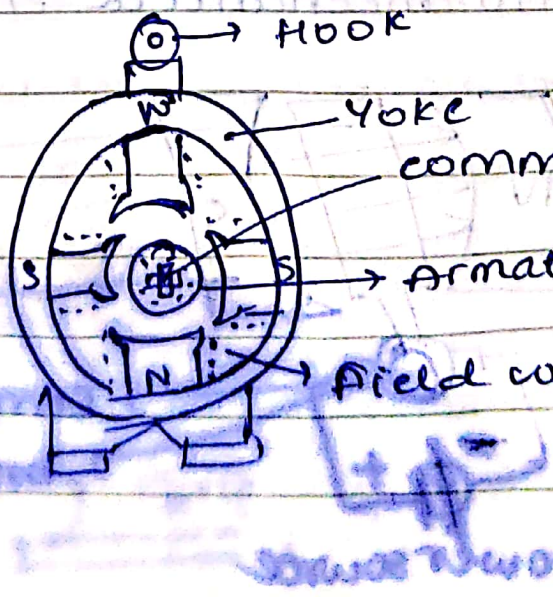
1) Explain the principle operator of D.C generator.

A) Principle:- A DC generator works on the principle of electromagnetic Induction It states that "when a conductor cuts the magnetic lines of force, an electromotive force (EMF) is induced in it. The directi of induced EMF is given by F-right Hand rule

Construction: Magnetic field system, Armature, commutator, Brushes.

Working:- When the armature rotates in the magnetic field, EMF is induced in its conductors. The commutator converts this alternating EMF into unidirectional voltage, & brushes collect the current from cummutator.

→ Thus a D.C generator converts Mechani-cal energy into electrical energy.



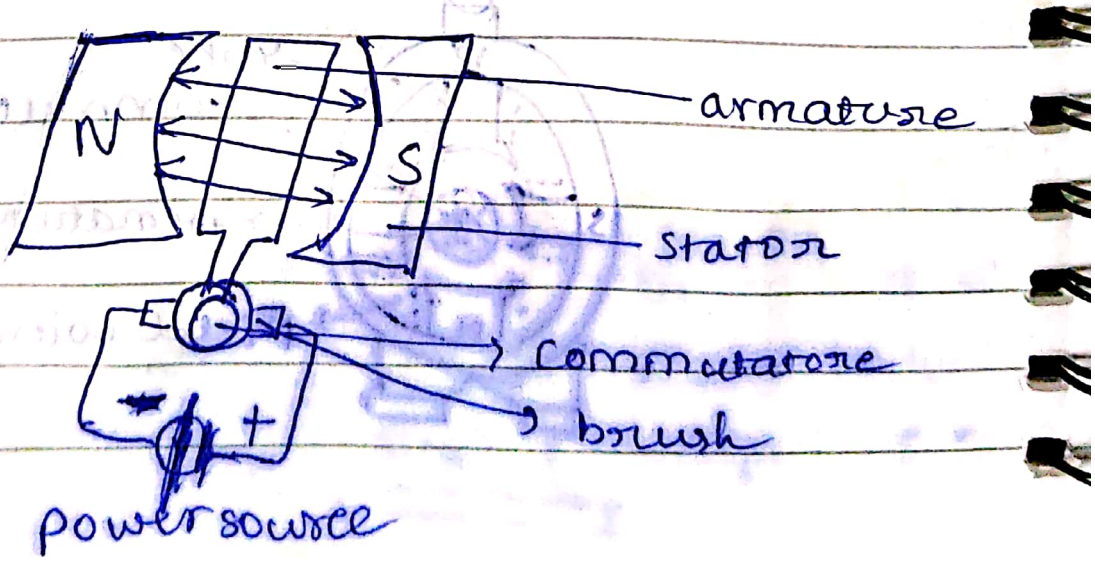
Q)-Explain the operation of D.C motor with neat diagram

A) Principle:- A DC motor works on the principle of EMF, which states that "when a current-carrying conductor is ~~parallel~~ placed in a magnetic field, it experiences a mechanical force". The direction of the force is given by F. Left hand rule

Construction:- Yoke, Field poles Armature core, commutator, Brushes, shaft.

Operation:- When current flows through the armature conductors placed in the field produced by field poles, a torque acts on the armature, causing it to rotate. The commutator reverses the current direction every half rotation to keep the motion continous in the same direction.

→ The DC motor converts electrical energy into mechanical energy.



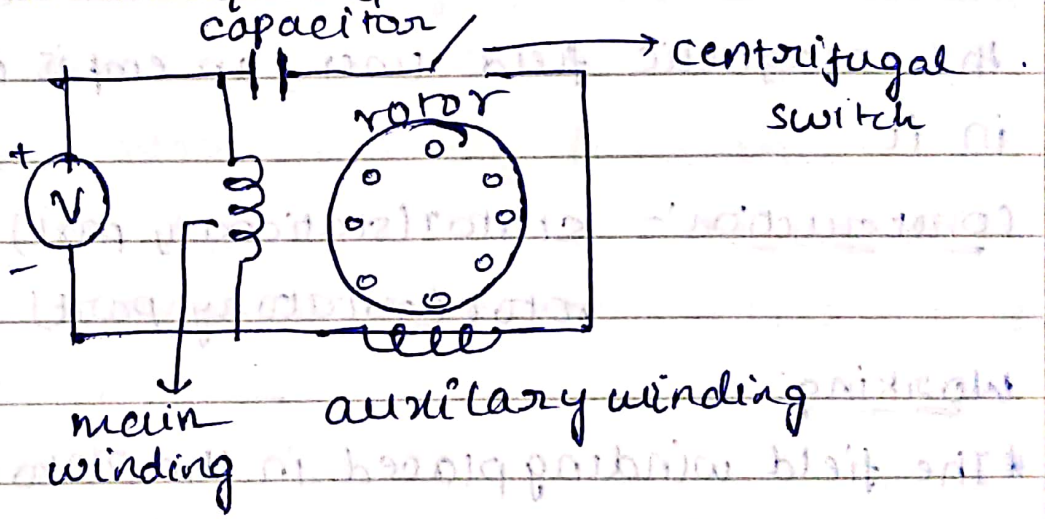
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5) Operation of Induction motor

a) single phase b) three phase.

A) a) 1. When a single-phase AC supply is given to the stator winding, it produces an alternating magnetic field.

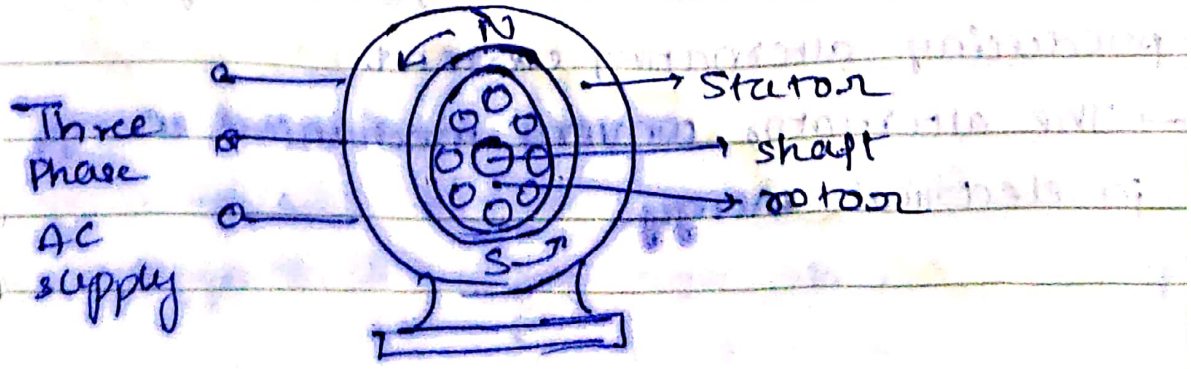
2. This field induces current in the rotor, which interacts with the stator field to produce torque & make the rotor rotate



b) 1. When a three phase AC supply is given to the stator, it produces a rotating magnetic field.

2. This RMF cuts the rotor conductors, inducing an EMF & current in them

3. The interaction b/w stator's rotating field and rotor current produces torque, causing the rotor to rotate in the direction of field



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4) Explain the operation of Alternator

Principle:-

A) An alternator is an electrical machine that converts mechanical energy into electrical energy.

* It works on the principle of electromagnetic induction
* It states that "When a conductor cuts the magnetic field lines, an emf is induced in it.

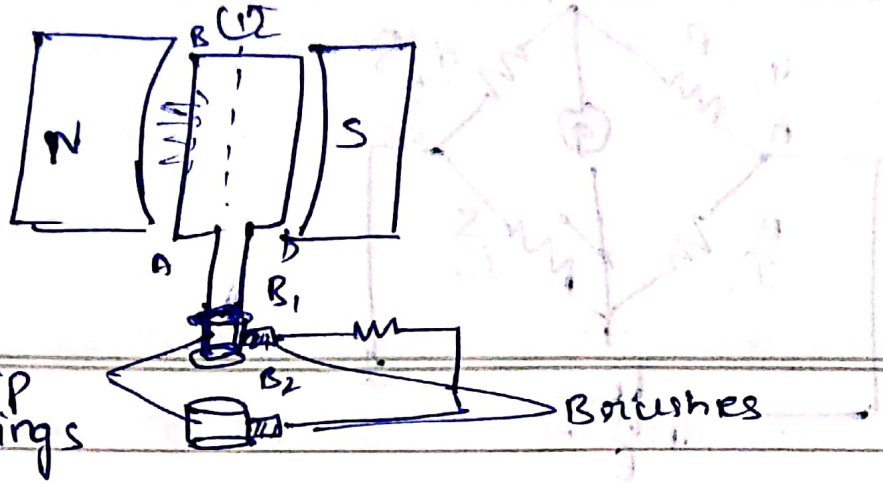
Construction:- stator (stationary part)
rotor (rotatory part)

Working:-

* The field winding placed in the rotor & armature winding placed in stator
* When the rotor is rotated by a prime mover, it produces a rotating magnetic field.
* This rotating fields cuts the stationary armature conductors, inducing an alternating EMF in them
* As the rotor continues to rotate, the direction of induced EMF reverses every half-cycle, producing alternating current.

→ The alternator converts mechanical energy in electrical energy.

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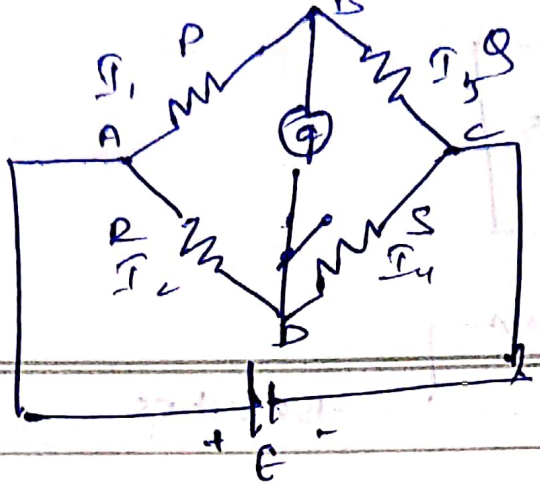
8) Explain the working of wheatstone bridge with neat diagram.

Principle:- When the bridge is balanced, the potential difference between the two mid points is zero, and no current flows through the galvanometer.

Working:-

1. A current flows through the circuit from the battery dividing into two parts.
2. The galvanometer detects only p.d b/w the midpoints of two paths.
3. When the bridge is unbalanced, the current flows through the galvanometer, indicating a P.D.
4. By adjusting the variable resistor P, the bridge can be balanced, the galvanometer shows no deflection
5. At this point using the relation $P/Q = R/S$

$$S = \frac{R \times Q}{P}$$



$$I_1 P = I_2 R \quad -(1)$$

$$I_1 = I_3 = \frac{E}{P+Q}$$

$$I_2 = I_4 = \frac{E}{R+S}$$

$$(1) \Rightarrow \frac{E}{P+Q} \cdot P = \frac{E}{R+S} \cdot R$$

$$P(R+S) = R(P+Q)$$

$$PR + PS = PR + RQ$$

$$PS = RQ$$

$$\boxed{S = \frac{RQ}{P}}$$

9) Explain the working of meggar with neat diagram.

A) Principle:- A meggar works on ohm's law, measuring resistance by comparing the current produced by a known voltage

Working:-

1. When the handle is rotated, the generator produces a dc voltage, which is applied

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across the insulation under test.

2. The current trough insulation depends on resistance

3. The galvanometer, shows the deflection corresponding to the insulation resistance on a calibrated scale.

4. A high resistance causes a large deflection and a low resistance causes a small deflection.

